CS 184 · SPRING 2026 · FINAL VIDEO

Limitless: Infinity

Repulsive surface optimization for collision-free shape paths. From canonical embeddings to a ball-through-tube demo.

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Collision-free motion, without collision detection.

The original Repulsive Shells paper asks for shape operations that do not pass through themselves. Instead of detecting collisions after they happen, it builds repulsion directly into the metric.

For the final project, we implemented the Repulsive Surfaces machinery underneath that idea, then used the shell/path layer for one focused interpolation demo.

An energy that becomes infinite at self-contact.

Our discrete tangent-point energy sums a kernel over triangle pairs. If two distant patches approach zero separation, the kernel diverges.

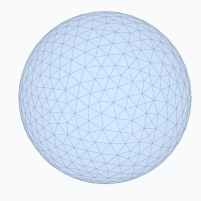
$$\widehat{\Phi}(x) = \sum_{t_1 \neq t_2} a_1 a_2 \frac{|\langle n_1, c_1 - c_2 \rangle|^6}{\|c_1 - c_2\|^{12}}$$

From paper formula to a working optimizer.

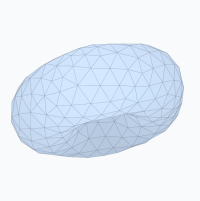
The pipeline computes face geometry and analytical gradients, accelerates TPE with a BVH and block-cluster tree, solves for Sobolev-preconditioned descent directions, and remeshes dynamically as the surface changes.

mesh → face geometry → TPE/BVH/BCT → preconditioned direction → line search → remesh

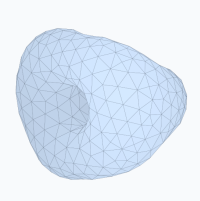
Canonical embeddings.



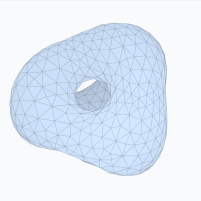
genus 0



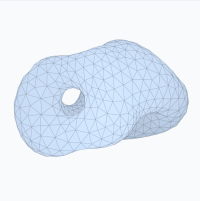
genus 1



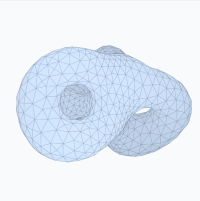
genus 2



genus 3

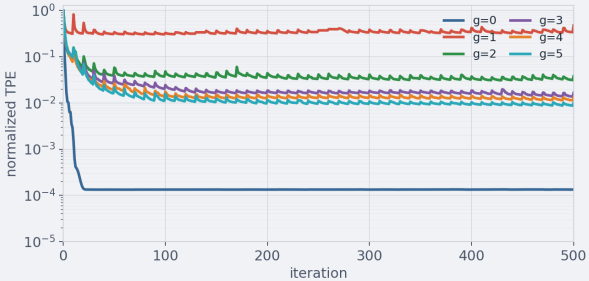


genus 4



genus 5

The energy drops at every genus.

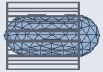


Six 500-iteration runs with dynamic remeshing.

Representative drops: genus 0 falls by 7473x; genus 5 falls by 99.6x.

Repulsor all-pairs parity stayed within floating-point tolerance on final meshes.

A sphere squeezes through a narrow tube.

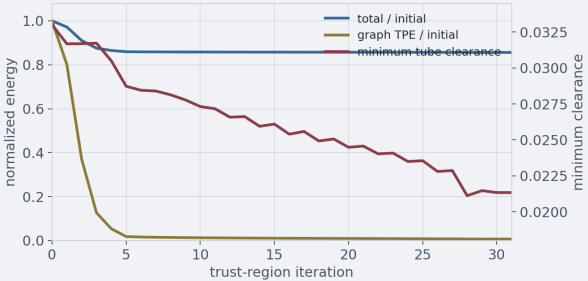


We optimized a 30-frame path with fixed endpoints.

The objective combines shell energy, self TPE, a fixed tube-surface TPE barrier, and a small SDF guard.

Final minimum tube clearance: 0.0213.

The hard part was numerical discipline.



Finite-difference gradient checks caught sign errors.

Repulsor parity caught kernel-bias mistakes that internal tests could not.

Deterministic reductions and remeshing budgets made outputs reproducible.

A working repulsive-surface pipeline.

We did not finish the full GPU Repulsive Shells system, but we built the CPU core it depends on: fast TPE, preconditioned descent, remeshing, validation, and a shape-path demo.

The final result is a reproducible system for generating embedded surfaces and collision-aware interpolation examples from saved mesh sequences.